

A. Deliverable title

AI Evolution and impact analysis: From data science to generative AI

B. Objective

The objective of this deliverable is for **you to conduct** a comprehensive analysis of the artificial intelligence evolution and its strategic implications within your educational and regional context. Through this project, **you will demonstrate** your understanding of AI's progression from data science through machine learning to generative AI, **analyze** current applications and trends, and **develop** strategic insights that align with educational priorities in Latin America.

C. Instructions

Please follow the instructions below to complete your deliverable.

1. **Download the Jupyter notebook:** Download the Jupyter notebook provided on the platform. This will be the space where you develop and document your analysis.
2. **Use the guide:** Review the sections and guiding questions provided below, as they are designed to help you structure your analysis. Based on this guide, develop and record your responses directly in your Jupyter notebook.
3. **Reflect on your context:** Throughout the activity, reflect on your own educational environment. Consider how the evolution of AI impacts your specific institutional and regional context, and incorporate these reflections into your analysis.

Reflection Guide

A. Current AI landscape in education

- Describe the current state of AI adoption in your educational environment and identify specific challenges or opportunities you observe.
- *Guiding question:* What specific educational challenges in your environment could benefit from understanding AI evolution and its applications?

B. AI Evolution understanding

- **Historical progression:** Analyze the evolution from data science to machine learning to generative AI and its impact on education.
- *Guiding question:* How has the progression from traditional data analysis to generative AI created new possibilities for educational innovation?
- **Transformer impact:** Examine how Transformer architecture and LLMs have changed the educational technology landscape.
- *Guiding question:* What educational applications have become possible due to advances in large language models and Transformer technology?

C. Strategic analysis objective

- State the main objective of your AI evolution analysis and what insights you hope to gain for your educational context.
- *Guiding question:* What specific insights about AI evolution do you hope to gain that will inform your educational practice and institutional strategy?

D. Analysis implementation plan

- **Research approach:** Detail the concrete steps you will take to analyze AI evolution and its educational implications.
- *Guiding question:* What specific research and analysis methods will you use to understand AI's impact on education?
- **Focus areas:** Explain which aspects of AI evolution (technical, educational, regional) you will prioritize and why.

- *Guiding question:* Which aspects of AI evolution are most relevant to your educational context and why?

E. Required resources and context

- **Information sources:** List the resources you need for your analysis, including technical documentation, case studies, and regional examples.
- *Guiding question:* What resources (research papers, case studies, expert insights) do you need to conduct a thorough AI evolution analysis?
- **Regional context:** Explain how you will incorporate Latin American educational context and examples into your analysis.
- *Guiding question:* How will you ensure your analysis reflects the specific opportunities and challenges in Latin American educational settings?

F. Analysis challenges

- **Knowledge gaps:** Identify potential obstacles in understanding complex AI concepts and technical developments.
- *Guiding question:* What aspects of AI evolution might be challenging to understand, and how will you address these knowledge gaps?
- **Contextual limitations:** Propose strategies to overcome limitations in accessing current information and regional examples.
- *Guiding question:* How will you ensure your analysis includes current, relevant information despite potential resource limitations?

G. Impact and application

- **Success indicators:** Define how you will know if your analysis has provided valuable insights for educational application.
- *Guiding question:* How will you measure whether your AI evolution analysis has generated actionable insights for your educational context?
- **Practical application:** Describe how you will apply insights from your analysis to real educational challenges and opportunities.
- *Guiding question:* How will you translate your understanding of AI evolution into concrete improvements in your educational practice?